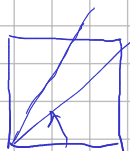
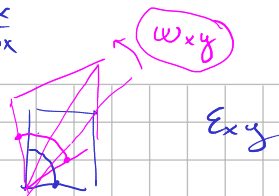


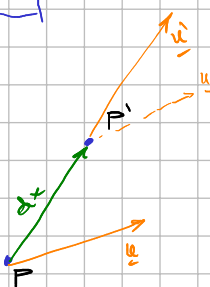
Rotación infinitesimal $\rightarrow$ desplazamiento $\underline{u}$
 Velocidad de spin $\rightarrow$ velocidad $\underline{v}$



$$\epsilon_{xx} = \frac{\partial u_x}{\partial x}$$



a)

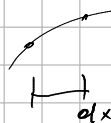


campo $\underline{u}(\underline{x})$

$$\underline{u} = \underline{u}(\underline{x}_P)$$

$$\underline{u}' = \underline{u}(\underline{x}_{P'})$$

$$\left. \begin{array}{l} \underline{u} = \underline{u}(\underline{x}_P) \\ \underline{u}' = \underline{u}(\underline{x}_{P'}) \end{array} \right\} \underline{u}' - \underline{u} = d\underline{u} = \underline{\nabla} \underline{u} \cdot d\underline{x}$$



$$(\underline{u}' - \underline{u})_i = du_i = (\underline{\nabla} \underline{u} \cdot d\underline{x})_i = \frac{\partial u_i}{\partial x_j} dx_j = \frac{z}{z} \frac{\partial u_i}{\partial x_j} dx_j = \frac{1}{2} \left(z \frac{\partial u_i}{\partial x_j} \right) dx_j =$$

$$= \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) dx_j$$

$$= \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \underbrace{\frac{\partial u_j}{\partial x_i} - \frac{\partial u_j}{\partial x_i}}_0 + \frac{\partial u_i}{\partial x_j} \right) dx_j$$

$$= \left[\underbrace{\frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)}_{E_{ij}} + \underbrace{\frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right)}_{-W_{ij}} \right] dx_j$$

$$= (E_{ij} - W_{ij}) dx_j$$

$\hookrightarrow$ simétrico $\rightarrow$ antisimétrico

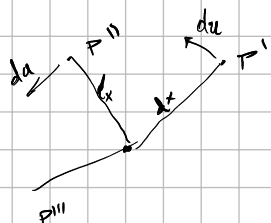
$$W_{ij} = -W_{ji} = -W_{ij}^T$$

$$\underline{u}' - \underline{u} = d\underline{u} = \underline{\nabla} \underline{u} \cdot d\underline{x} = (\underline{E} - \underline{W}) d\underline{x}$$

Assumamos un movimiento sin deformaciones (cuerpo rígido). $\Rightarrow E = 0$

$$\underline{u}' - \underline{u} = d\underline{u} = -\underline{W} d\underline{x}$$

$$du_i = -W_{ij} dx_j$$



$$W_{ij} = \epsilon_{ijk} \omega_k$$

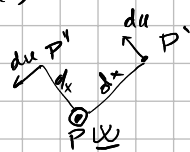
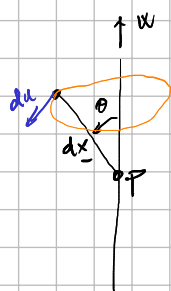
$$R_{ij} = \frac{1}{2} \epsilon_{ijk} R_k$$

$$du_i = -\epsilon_{ijk} \omega_k dx_j = \epsilon_{ikj} \omega_k dx_j \Rightarrow d\underline{u} = \underline{\omega} \times d\underline{x}$$

$$|du| = |\underline{\omega}| |dx| \sin(\theta)$$

producto cruz para llegar aca

$$\omega \perp dx \Rightarrow \theta = \pi/2 \Rightarrow \sin(\theta) = 1 \Rightarrow |du| = |\underline{\omega}| |dx|$$



b)

$$\omega_k = \frac{1}{2} \varepsilon_{kij} \omega_{ij} \rightarrow \omega_{ij} = \varepsilon_{ijk} \omega_k$$

$$\omega_k = \frac{1}{2} \varepsilon_{kij} \omega_{ij} \times \varepsilon_{mnk}$$

$$\varepsilon_{mnk} \omega_k = \frac{1}{2} \varepsilon_{kij} \varepsilon_{mnk} \omega_{ij}$$

$$= \frac{1}{2} (\delta_{in} \delta_{jm} - \delta_{im} \delta_{jn}) \omega_{ij}$$

$$= \frac{1}{2} (\omega_{mn} - \omega_{nm})$$

$$\omega_{nm} = -\omega_{mn}$$

$$= \frac{1}{2} (2 \omega_{mn}) = \omega_{mn} \Rightarrow$$

$$\omega_{mn} = \varepsilon_{mnk} \omega_k$$

$$\omega_{ij} = \varepsilon_{ijk} \omega_k$$

c) es equivalente al realizado previamente

$$\underline{\Omega} = \text{rot}(\underline{v}) = \nabla \times \underline{v}$$

$$\Omega_{ij} = \frac{1}{2} \left(\frac{\partial v_j}{\partial x_i} - \frac{\partial v_i}{\partial x_j} \right)$$

$$\Omega_k = \varepsilon_{kij} \Omega_{ij} = \varepsilon_{kij} \frac{1}{2} \left(\frac{\partial v_j}{\partial x_i} - \frac{\partial v_i}{\partial x_j} \right) =$$

$$= \frac{1}{2} \left(\varepsilon_{kij} \frac{\partial v_j}{\partial x_i} - \varepsilon_{kij} \frac{\partial v_i}{\partial x_j} \right) = \varepsilon_{kij} \frac{\partial v_j}{\partial x_i} = (\nabla \times \underline{v})_k$$

$$\Omega_k = (\nabla \times \underline{v})_k \Rightarrow \underline{\Omega} = \nabla \times \underline{v}$$